

Codebook

Wind Turbine and Election Data Set and Repository

September 1, 2026

1 Description of Data Files in the Repository

The repository contains the following data sets and processing scripts to construct the respective data sets:

File Name	Description
<code>construct_df_panel.R</code>	Combines GERDA election data, wind turbine counts, and covariates into the final panel datasets (<code>df_panel_cov.rds</code> , <code>df_panel_wo_cov.rds</code> and <code>df_did_ready.rds</code>).
<code>construct_df_wind_count.R</code>	Constructs the wind turbine count data frame (<code>df_wind_count.rds</code>) and performs spatial joins between turbine GPS coordinates and municipality shapefiles across all election periods.
<code>construct_df_cov.R</code>	Constructs <code>df_covariates.rds</code> by combining municipality- and county-level covariates from the INKAR database into a single panel dataset.

File Name	Description
<code>df_covariates.rds</code>	Dataset covering the period 1995–2023 with 19 socio-economic variables at county and municipality level.
<code>df_did_ready.rds</code>	Same panel dataset as the main dataset <code>df_panel_cov.rds</code> but including treatment variables as required by different packages as well as the constructed outcome <code>current_incumbent</code> .
<code>df_panel_cov.rds</code>	Panel dataset covering the period 1990–2025 with federal election results on municipality level, wind power plant counts by election year, and 19 socio-economic covariates.
<code>df_panel_wo_cov.rds</code>	Panel dataset covering the period 1990–2025 with federal election results on municipality level and wind power plant counts by election year (without additional covariates).
<code>df_wind_count.rds</code>	Intermediate data frame of wind turbine counts per municipality and election period, constructed by <code>construct_df_wind_count.R</code> .

2 Description of Variables in the Main Data Set

The table below documents all variables present in `df_panel_cov.rds`.

Variable	Description
<code>ags</code>	Unique municipality identifier (Amtlicher Gemeindegemeinschaftsschlüssel).

Variable	Description
ags_name	Name of the municipality.
election_year	Year of elections.
county	County identifying number.
turnout	Voter turnout as a proportion of eligible voters.
cdu	Vote share for CDU (Christian Democratic Union).
csu	Vote share for CSU (Christian Social Union).
spd	Vote share for SPD (Social Democratic Party).
fdp	Vote share for FDP (Free Democratic Party).
linke_pds	Vote share for Die Linke/PDS (Left Party).
gruene	Vote share for Bündnis 90/Die Grünen (The Greens).
afd	Vote share for AfD (Alternative for Germany).
cdu_csu	Combined vote share for CDU and CSU.
far_right	Vote share for far-right parties.
far_left	Vote share for far-left parties.
far_left_w_linke	Vote share for far-left parties including Die Linke.
freie_wahler	Vote share for Freie Wähler (Free Voters).
population	Total population of the municipality.
pop_density	Population density of the municipality.
wind_count_0km	Number of new wind turbines (with 0km buffer).

Variable	Description
wind_count_1km	Number of new wind turbines (with 1km buffer).
wind_count_3km	Number of new wind turbines (with 3km buffer).
wind_count_5km	Number of new wind turbines (with 5km buffer).
wind_count_10km	Number of new wind turbines (with 10km buffer).
cum_wind_count_3km	Cumulative number of wind turbines (with 3km buffer).
lag_wind_count_3km	Lagged count of new wind turbines (with 3km buffer).
cum_lag_wind_count_3km	Lagged cumulative number of turbines (with 3km buffer).
wind_count_muni	Number of new wind turbines within the municipality.
lag_wind_count_muni	Lagged count of new wind turbines within the municipality.
cum_wind_count_muni	Cumulative number of wind turbines within the municipality.
cum_lag_wind_count_muni	Lagged cumulative number of turbines within the municipality.
dummy_lag_wind_count_3km	Dummy variable indicating the presence of lagged wind turbines (with 3km buffer).
cat_cum_wind_count_3km	Categorical cumulative wind turbine count (grouped in bins of 10, with 3km buffer).
treat_nonabsorbing	Treatment defined as non-absorbing – 1 if new turbines occur in the period, 0 if not (3km buffer).
treat_absorbing	Treatment defined as absorbing – 1 starting from the period were first turbine was built (3km buffer).

Variable	Description
treat_absorbing_cs	Treatment defined as absorbing; in the cohort-format needed by Callaway & Sant'Anna.
average_age	Average age of the population.
pop_fem	Total female population.
unemp	Number of unemployed individuals.
inc_tax	Income tax revenue.
busi_tax	Business tax revenue.
tax_rev	Total tax revenue.
agri_land	Total area of agricultural land.
commute_balance	Net commuter balance.
dens_work	Resident-to-job density.
employees	Employees subject to social security contributions at their place of residence.
net_migration	Total net migration balance.
purch_pow	Purchasing power.
tourism	Number of overnight tourist stays in accommodation establishments per capita.
hinc	Average monthly household income.
share_foreign	Share of foreign population.

Variable	Description
<code>educ</code>	Share of employed individuals at their place of residence with a vocational qualification.
<code>share_fem</code>	Share of females in the total population.
<code>share_unemp</code>	Unemployment rate (share of unemployed individuals).
<code>share_emp</code>	Share of employees subject to social security contributions in the population.

3 Notes on Reproducibility

3.1 Open Code

All code, the codebook and the final data sets required to reproduce the main results are available on GitHub at <https://github.com/wind-power>. The repository includes all pre-processing scripts and the main analysis code.

3.2 Data Availability

Election data. Federal election results at the municipality level were downloaded from the German Election Database (GERDA), an independent research project providing harmonised election data. The dataset used is `federal_muni_harm_25` (2025 municipality borders, downloaded June 1 2026). For details on GERDA see Heddesheimer et al., 2025.

Wind power plants and spatial data. Wind turbine data, including GPS coordinates, were downloaded from the Central Market Registry (Marktstammdatenregister, n.d.). The applied filters were: commissioning date before 23.02.2025; hub height greater than 50 m; operational status was not "In Planung"; wind type "Wind an Land"; non-missing latitude

and longitude; energy carrier "Wind". Shapefiles for the spatial analysis are the VG250 (Verwaltungsgebiete 1:250 000, reference date 01.01.2025), downloaded from the Federal Agency for Cartography and Geodesy (Bundesamt für Kartographie und Geodäsie, n.d.). Note that both the election data and the wind turbine spatial assignment use 2025 municipality boundaries as the reference standard.

Covariate data. All covariate data (except population and population density) were obtained from the INKAR database (BBSR, n.d.) of the Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR). Because INKAR uses municipality boundaries harmonised to 2023 while election and wind-power data use 2025 boundaries, some municipalities merged or split between the two reference years cannot be matched; see Section 3.3 for details. INKAR provides data only for the period 1995 to 2023, and some indicators are available exclusively at the county level. When incorporating INKAR covariates, the year 1990 is excluded from the analysis due to missing data. For 1994, values from 1995 are used as imputations, while for 2025 the most recent available data from 2023 is applied to retain these periods.

The table below provides an overview of all covariates, their temporal availability, and the german short name for exact identification in the INKAR database:

Covariate	Variable Name	German Name & Availability	Level
Average age	<code>average_age</code>	Durchschnittsalter der Bevölkerung (2002–2023)	Municipality
Population female	<code>pop_fem</code>	Bevölkerung weiblich (1995–2023)	Municipality
Income tax	<code>inc_tax</code>	Einkommenssteuer (1995–2023)	Municipality
Business tax	<code>busi_tax</code>	Gewerbesteuer (1995–2023)	Municipality

Covariate	Variable Name	German Name & Availability	Level
Tax revenue	<code>tax_rev</code>	Steuereinnahmen (1995–2023)	Municipality
Employees subject to social security	<code>employees</code>	Sozialversicherungspflichtig Beschäftigte am Wohnort (1998–2023)	Municipality
Workplace density	<code>dens_work</code>	Einwohner-Arbeitsplatz- Dichte (2000–2023)	Municipality
Number of unemployed	<code>unemp</code>	Arbeitslose (1998–2023)	Municipality
Commuter balance	<code>commute_balance</code>	Pendlersaldo (1998–2023)	Municipality
Overnight tourist stays	<code>tourism</code>	Gästeübernachtungen in Beherbergungsbetrieben (2009–2023)	Municipality
Net migration	<code>net_migration</code>	Gesamtwanderungssaldo (1995–2023)	Municipality
Purchasing power	<code>purch_pow</code>	Kaufkraft (2013–2023)	Municipality
Agricultural land	<code>agri_land</code>	Landwirtschaftsfläche (2017–2023)	Municipality
Average monthly household income	<code>hinc</code>	Monatliches Haushaltseinkommen (2000–2022)	County
Share of foreign population	<code>share_foreign</code>	Ausländeranteil (1995–2023)	County

Covariate	Variable Name	German Name & Availability	Level
Working population with vocational qualifications	educ	Beschäftigte am WO mit Berufsabschluss (2012–2023)	County

3.3 Unresolved Merge Cases

Three distinct matching issues arise when assembling the final panel from the component data sources. None of these affect municipalities for which both election results and wind turbine counts exist.

Wind turbine municipality specification (robustness check only). When the no-buffer robustness specification assigns turbines to municipalities via their official municipality key (AGS) rather than GPS coordinates (i.e. there is no spatial buffer), 23 municipalities in the wind registry cannot be matched to any municipality in the spatial panel — either by AGS or by name. As a consequence, 75 of 28,630 turbines (0.26%) are excluded from this specification. This issue does not affect the main 3 km buffer specification, which relies solely on GPS coordinates.

Municipalities in `df_wind_count` not in `df_panel_wo_cov`. After joining the wind turbine count data to the GERDA election data, 205 municipalities present in the spatial panel (`df_wind_count`) are dropped because they have no corresponding entry in the election data. Of these, 204 are uninhabited areas or nature reserves; one municipality, Wiedenborstel, has fewer than 10 registered inhabitants. These observations are not relevant to the analysis.

Municipalities in `df_covariates` not in `df_panel_wo_cov`. When merging the INKAR covariate data into the main panel via the AGS, 229 municipalities present in the covariate file are dropped. Of these, 204 are the same uninhabited or near-uninhabited units described above, one is Wiedenborstel and three could be subsequently matched via names. The

remaining 21 municipalities were merged with or split from neighbouring units between the 2023 INKAR reference date and the 2025 election/wind-data reference standard; crosswalks for these units are not yet available from the INKAR website. No municipality for which both election results and wind turbine counts exist loses its covariate information as a result of this mismatch, i.e. this will not affect final sample size.